Strongly Polynomial Near-Linear Preprocessing for All-Target Restricted Shortest Paths

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Given a directed graph with nonnegative lengths and costs, a source, and a cost budget, restricted shortest paths asks for a shortest budget-feasible path to every target. We give a deterministic bicriteria approximation with near-linear preprocessing whose arithmetic-operation count is independent of all numerical magnitudes, aspect ratios, and input precision. For accuracy ε, the representation reports a finite length bound and recovers an actual path of cost at most $(1+\varepsilon)C$ and length at most $(1+\varepsilon)\,\mathrm{OPT}_C(t)$ for every feasible target. The construction combines a stopped two-clock boundary oracle with clipped offline activation and edge-local scale events. Whole-vertex activation and positive failure prices allow one first-hit approximation factor per recursive level. Finite-rank policies control rounding error, and conservative dyadic clock laws and bounded-space deterministic decoding implement the construction without exponential or arbitrary-weight logarithm oracles. The parameter dependence is deliberately loose; the result concerns strong polynomiality rather than practical efficiency.

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1 The result and its context

A single-target reduction can remove numerical-aspect dependence by locating a feasible bottleneck edge and searching an interval of ratio n. Doing this separately for every target introduces an unwanted factor of n. The difficulty addressed here is sharing the work over all target-specific scales.

Ashvinkumar's deterministic clock algorithm [1], Theorem 1, gives all-target bicriteria restricted shortest paths (RSP) in $\widetilde{O}(m \log(nW)/\varepsilon^9)$ time, where W is the positive-length aspect ratio. The paragraph after its Theorem 2 explicitly distinguishes the strongly polynomial all-target question from the single-target consequence. The earlier work of Ashvinkumar, Bernstein, and Karczmarz [2], Lemma C.1 and the open-problem discussion, makes the same distinction. We remove the numerical-aspect factor with a larger polynomial dependence on accuracy.

Theorem 1. *There are an absolute K and a uniform deterministic algorithm with the following guarantee. Let G be a directed adjacency-list graph with $n \geq 2$ labeled vertices, m edges, arbitrary finite nonnegative real lengths ℓ and costs c, source s, budget $C > 0$, and rational $0 < \varepsilon \leq 1/2$. In at most*

$$K(m+n)\log(n+2)^{20}\varepsilon^{-100}$$

elementary operations and space, the algorithm constructs an all-target representation. Every target admitting a path of cost at most C is represented by a finite bound B_t and compact witness data. A deterministic query recovers an actual path P_t satisfying

$$c(P_t) \leq (1+\varepsilon)C, \qquad \ell(P_t) \leq B_t \leq (1+\varepsilon)\,\mathrm{OPT}_C(t).$$

A target without a C-feasible path may be unavailable or represented by an actual path satisfying the relaxed cost bound. Unreachable targets have no fabricated witness. Queries take polynomial time in $n, \log(n+2), 1/\varepsilon$ plus output size, independently of numerical aspects, and use space within the displayed bound.

The model charges exact real $+, -, \times, /$, comparisons, and ordinary graph, control, and index operations. There is no primitive logarithm or floor of an arbitrary input weight, exponential evaluation, random sampling, or policy-synthesis oracle. For rational inputs, all intermediate encodings have polynomial bit length in the input encoding and $n, m, \log(1/\varepsilon)$. Arithmetic-operation bounds are not bit-time bounds for reading arbitrarily long numbers. Control indices have $O(\log(n+m+1/\varepsilon))$ bits up to an absolute factor. Parallel edges, zero resources, disconnected graphs, equal weights, and zero optima are included. Nonnegative walks may be cycle-erased.

The clock dynamic program and conditional-expectation witness potential originate in [1], Sections 4 and 6. Recursive offline activation and multiplicative counting have ordinary-SSSP precedents in Górkiewicz and Karczmarz [3], Section 6, and McCauley, Moseley, Niaparast, Niaparast, and Singh [4], Theorem 1. Those additive-distance results do not provide a stopped stochastic boundary contract. Our additional ingredient is a costed nonadditive oracle compatible with whole-vertex activation, clipping, and finite-precision error; local events then cover all scales without a numerical-aspect scan.

Research methodology. OpenAI Codex materially assisted the algorithm and proof development, diagnostic computations, literature checks, and exposition throughout this paper. The end-of-paper AI methodology statement details these roles.

2 A stopped clock oracle

Fix an integer $q \geq 2$, $\rho > 0$, and $F = 1 + \rho$. A state is (v, i, j), $0 \leq i, j < q$. An original edge uv is a backwards action from v to u adding nonnegative integer increments (a, b) drawn from a finite law p_e. Either coordinate reaching q pays F; source states immediately pay ρ. Nontermination also pays F. A

voluntary tick increments either coordinate by one. Let X be the infimum expected payment. Thus $X = \rho+$ optimal failure probability. Extra initial ticks cannot help: simulate the same physical actions while tracking virtual larger clocks. Hence X is nondecreasing in i, j and belongs to $[\rho, F]$.

Activate a set A of whole vertices excluding s. Every inactive vertex is terminal with a clock-monotone table $B_v \in [\rho, F]$, and the source remains exact. A nonmonotone supplied upper bound can be replaced by its upper-orthant minimum in $O(q^2)$ operations per encountered vertex. If $X \le B \le \beta X$, this closure retains both inequalities, since X is clock-monotone.

Lemma 2. *Assume $X \le B \le \beta X$. The stopped optimum Y on A is computable using only incident edges in*

$$O((|A| + d_A)q^4 \log(|A| + 2) + (|A| + d_A)q^2 H)$$

operations, where d_A counts adjacency incidences at active vertices and H is a boundary-coordinate access cost. Moreover $X \le Y \le \beta X$. It has an optimal pointer policy terminating on every trajectory within $(2q+1)(|A|+1)$ decisions.

Proof. Process clock layers in decreasing $i + j$, with any order within a sum. At (i, j), seed each active vertex with its smaller one-tick value, using F at overflow, and with the expected payoff of every edge having an inactive predecessor. Maintain a min-heap on active vertices. On popping u, fix its value $Z(u, i, j)$ and relax every edge uv whose head is still unsettled, using

$$p_{\text{over}} F + \sum_{a<q-i,\, b<q-j} p_e(a, b) Z(u, i + a, j + b). \tag{1}$$

Store the improving seed or action; ties keep the existing choice. Positive-increment terms are already known. Initial tick seeds imply that every higher-clock value at u is at least $Z(u, i, j)$. Thus a relaxed candidate is a convex combination of values at least the popped value, including F, and heap removals are nondecreasing.

For each action the final table satisfies $Z \le \mathbb{E} Z_{\text{next}}$. Inactive predecessors were seeded; earlier-settled active predecessors were relaxed. For a later-settled predecessor, every term is at least its popped value and hence at least the current value. Self-loops have the same property. Tick inequalities follow from seeding. Iterating these inequalities for any policy, charging F to trajectories still active after k steps, gives an expected payment at least Z. Bounded convergence, including the stipulated payment on nontermination, gives $Z \le Y$.

The chosen policy realizes Z. A zero-increment chosen edge points to an earlier pop rank at the same layer. A positive increment or a tick increases $i + j$. There are fewer than $2q$ unit advances before overflow, so every trajectory terminates within $(2q + 1)(|A| + 1)$ decisions. Backwards substitution gives expected payoff exactly Z, proving $Y = Z$. In particular a closed zero-increment class receives F through tick seeds rather than an artificial smaller Bellman fixed point.

For the comparison, replace the first inactive terminal by a nearly optimal full continuation to obtain $X \le Y$. Conversely stop a nearly optimal full policy at its first inactive visit. Its conditional continuation payment dominates X, whereas $B \le \beta X$. Source, overflow, and nontermination payments are at most β times themselves. Taking limits gives $Y \le \beta X$, with one factor per trajectory.

Build both adjacency directions once. A call enumerates active vertices and only their incidences; inactive tables are fetched and cached only at encountered predecessors. Each of q^2 layers has a heap and $O(q^2)$ work per edge expectation. Timestamps or local dictionaries avoid clearing a global array. Local space is $O((|A| + d_A)q^2)$ apart from external tables. □

For a dyadic mesh h_0, round every edge candidate upward before comparison. Assume terminal values and ρ, F are on the mesh. Causality, tick monotonicity, and the chosen-policy rank remain valid. Put

$N = (2q+1)(n+1)$. The rounded table satisfies

$$Y \le Z \le Y + Nh_0. \tag{2}$$

Indeed the selected action has $\mathbb{E}Z_{\text{next}} \le Z$. Every action has $Z \le \mathbb{E}Z_{\text{next}} + h_0$: for a later predecessor the stronger unrounded inequality holds. Apply this residual inequality along the exact oracle's finite-rank optimal policy, of horizon at most N. This proves the upper bound without an absorption-time assumption on arbitrary policies.

3 Clipped offline activation

Suppose there are E ordered snapshots, with edge increments stochastically nonincreasing in time. A monotone coupling and virtual-clock simulation imply that exact full values X_t are nonincreasing in time. Fix

$$D = \max(1, \lceil \log_2 \max(1, E-1) \rceil), \quad \zeta = \frac{1}{16(2D+1)}, \quad \alpha = 1 + \zeta,$$

and dyadic $h_0 \le \min(\zeta\rho/N, 1/(2^{20}N))$. Integer logarithms here and below are computed by bounded doubling.

Solve the two endpoints with the rounded oracle. Set $U_{E-1} = Z_{E-1}$ and $U_0 = \max(Z_0, Z_{E-1})$ coordinatewise; both lie between X and αX and are ordered. For an interval $[l, r]$ with an interior midpoint t, activate v exactly when some clock entry satisfies $U_l(v) > \alpha U_r(v)$. At the midpoint use U_l as the inactive boundary, solve active vertices by the rounded oracle, and set

$$U_t = \max\{U_r, \min\{U_l, Z\}\} \tag{3}$$

on active entries, retaining U_l on inactive entries. Recurse on both halves. For one or two snapshots only endpoint solves are needed.

Lemma 3. *Every stored estimate is clock-monotone, globally nonincreasing in time, and satisfies*

$$X_t \le U_t \le \alpha^{2D+1} X_t \le (16/15) X_t. \tag{4}$$

Each vertex is active at $O(q^2D^2)$ calls. With M supplied edges, per-edge snapshot lookup cost H, and $h_0 = 2^{-b_0}$, total work, excluding coefficient preparation, is

$$O\big(E + (M+n)(1+q^2D^2)q^4[H + \log(n+2) + b_0]\big), \tag{5}$$

Proof. At depth d, suppose endpoints have factor $\beta_d = \alpha^{2d+1}$. For an inactive entry, $X_t \le U_l \le \alpha U_r \le \alpha\beta_d X_t$. The first-hit lemma and (2) therefore bound the raw midpoint by $X_t \le Z \le \alpha^2\beta_d X_t$. Clipping preserves this bound because both endpoints dominate X_t and $U_r \le \beta_d X_t$. Minima and maxima preserve clock monotonicity, and $U_l \ge U_t \ge U_r$. Induction proves (4); the last inequality follows by bounding the binomial expansion by the geometric series of ratio $1/16$. Clipped labels are estimates, not realizing policies.

A vertex inactive at a parent remains inactive in every descendant, since the earlier endpoint only decreases and the later endpoint only increases. Test only parent-active vertices in child calls, after the root scan. At a fixed depth, intervals have disjoint interiors. For each vertex-clock entry, time monotonicity telescopes the product of endpoint ratios to at most F/ρ. Thus this entry witnesses at most $\log(F/\rho)/\log\alpha$ active intervals. Union over q^2 entries and D depths gives $O(q^2D^2)$ calls for fixed ρ. The same bound weighted by adjacency degree charges incident-edge scans, including parallel edges.

Store each q^2 vertex block as a leaf in a persistent balanced tree on vertex labels. Access costs $O(\log(n+2))$; updating a block copies $O(\log(n+2))$ nodes and its q^2 entries. At a midpoint inherit the earlier version and update only active blocks. Cache boundary blocks at incident endpoints and reuse static scratch depth first. This proves (5), including bounded-comparison rounding. Persistent space is $O(nq^2 + E + nq^2D^2[q^2 +$

$\log(n+2)]$), plus graph, coefficient, and incident-boundary scratch. A root pointer per snapshot allows binary search for a target's first threshold crossing in $O(\log(E+2)\log(n+2))$ accesses. □

4 Finite clock laws and local scale events

Throughout the remainder set

$$\eta = \varepsilon/1000, \quad \rho = 1/32, \quad C' = (1+2\eta)C, \quad q = 2^{\lceil \log_2(2^{20}\eta^{-2})\rceil}, \quad U = 4q. \tag{6}$$

All rounded integers are located by binary comparisons on an explicitly bounded interval. Thus rounding on a mesh $h = 2^{-b}$ inside $[0, A]$ costs $O(b + \log(A+2))$ operations; this is not a floor primitive on unbounded inputs.

Lemma 4. *For a dyadic mean $0 < \mu \le U$, there is a finite dyadic law on $0, \dots, q$ (q denotes overflow) whose CDF $L_\mu(j)$ satisfies*

$$0 \le \mathbb{P}[\mathrm{Pois}(\mu) \le j] - L_\mu(j) \le \tau \quad (0 \le j < q), \qquad \tau \le \frac{1}{2^{20}Nq}.$$

For an increasing list of means these laws can be made stochastically increasing. Their distance in total variation from truncated Poisson is at most $q\tau$, and for $0 < z < 1$,

$$\sum_{k<q} p_\mu(k) z^k \le \exp((z-1)\mu). \tag{7}$$

Construction uses polynomially bounded dyadic precision and finite arithmetic, without evaluating an exponential.

Proof. Put $r = 1 + \lceil \log_2(n+2)\rceil + \lceil \log_2(q+2)\rceil$, $b = 64r$, $h = 2^{-b}$, and $H = 2U + b + 2$. Locate $k_0 = \lfloor \mu \rfloor$ by comparisons against integers in $[0, U]$. Relative Poisson weights $R(k) = \mathbb{P}[\mathrm{Pois}(\mu) = k]/\mathbb{P}[\mathrm{Pois}(\mu) = k_0]$ obey $R(k_0) = 1$, downward recurrence $R(k-1) = R(k)k/\mu$, and upward recurrence $R(k+1) = R(k)\mu/(k+1)$. Every multiplier away from the mode is at most one. Compute these recurrences through $0, \dots, H$, rounding downward to h after each step, to obtain $w(k)$. Then $0 \le R(k) - w(k) \le Hh$. Beyond $2U$, ratios are at most $1/2$, so the omitted tail after H is at most h.

For $S = \sum_{k\ge 0} R(k) \ge 1$, define

$$S_- = \sum_{k=0}^{H} w(k), \qquad A = ((H+1)^2 + 1)h, \qquad S_+ = S_- + A.$$

Then $S_- \le S \le S_+$. Set $L_\mu(j) = \lfloor \sum_{k\le j} w(k)/S_+ \rfloor_h$. Numerator deficit, normalization error, and final rounding give CDF deficit at most $\tau = 2A + h$. Since $H + 1 \le 2^{r+7}$ and $Nq \le 2^{3r+3}$, $b = 64r$ implies the displayed bound on τ.

For means $\mu_1 < \cdots < \mu_J$, replace each CDF by its prefix minimum over mean indices. Each earlier true CDF dominates the current one, so the error bound survives. CDFs remain increasing in j and decreasing in mean. Their differences, with the remaining mass at overflow, are normalized nonnegative dyadics. Telescoping the CDF deficits bounds total variation by $q\tau$. Stochastic dominance applied to the decreasing function z^k for $k < q$ and zero at overflow proves (7); its ideal expectation is at most the full Poisson generating function. At mean zero use the exact point mass at zero.

Per mean, work is $O((H+q)(b+\log(H+2)))$ and storage is $O(q)$. Recurrence temporaries have $O(b + \mathrm{bits}(\mu) + \log(H+2))$ bits, and dyadic summation adds only logarithmic precision. Exact long products of relative weights are never stored. □

Choose dyadic $\delta \in (\eta/(200n), \eta/(100n)]$ and mesh $h_\mu \le \eta\delta/100$. Generate means from δ by multiplying by $1 + \eta/4$ and rounding upward to h_μ, clipping the last point to U before bounded rounding. Successive ratios are at most $1 + \eta/2$ and, except possibly the last, at least $1 + \eta/4$. There are $J = O(\eta^{-1} \log(nq/\eta))$ means, with $O(\log(nq/\eta))$-bit encodings. Zero stays zero; a positive raw mean rounds upward to the first grid point, or is blocked above U. For an unblocked raw mean x,

$$x \le \widehat{x} \le (1 + \eta/2)x + \delta. \tag{8}$$

Compile these means using Lemma 4.

For an edge of positive length a, length-clock changes occur precisely at

$$L = \frac{qa}{(1 + 4\eta)\mu_j}. \tag{9}$$

At equality the new smaller mean applies; below its first event the edge is blocked, and after its last event it uses δ. Zero length uses mean zero. Sort the union of these finite edge-local lists and merge equal events. Each edge keeps its global event indices, so its law at a snapshot is found by predecessor search in $O(\log(J + 2))$ comparisons. Laws decrease stochastically in time. No global exponent of an input weight is computed.

5 Preprocessing and completeness

First delete edges with $c(e) > C$. Bucket normalized costs $c(e)/C$ into zero, a tiny bucket $(0, \delta_c]$ with dyadic $\delta_c \in (\eta/(2n), \eta/n]$, and a rounded geometric grid to 1. Use the preceding grid procedure with starting point δ_c, factor $1 + \eta/4$, and mesh at most $\eta\delta_c/100$. The $J_c = O(\eta^{-1} \log(n/\eta))$ positive buckets have ratios at most $1 + \eta/2$. For each ordered endpoint pair and bucket retain a minimum-length original edge, breaking ties by input order. A balanced map on at most $n^2(J_c + 1)$ keys processes arbitrary m in $O(m[\log(n + 2) + \log(J_c + 2)])$ operations, avoiding an $m \log m$ cost. Let $M \le \min(m, n^2(J_c + 1))$ denote the retained count.

Each edge has a representative e' with

$$\ell(e') \le \ell(e), \qquad c(e') \le (1 + \eta)c(e) + \eta C/n. \tag{10}$$

For a simple C-feasible path, replacement preserves its vertex sequence and gives cost at most C'. Nonnegative cycles may be removed, so optimum paths can be chosen simple. Original edge identifiers are retained for output.

Run ordinary cost SSSP on retained zero-length edges. Every vertex with distance at most C' gets bound zero and an SSSP-tree witness. This covers every original zero optimum. Standard label-setting tie rules choose a tree even with zero cycles. The source gets its empty path.

For all remaining targets, use raw cost mean $qc(e)/((1 + 4\eta)C')$ and raw length mean $q\ell(e)/((1 + 4\eta)L)$. Round upward and compile independent clocks as above. There are $E \le MJ$ merged events. If $E = 0$, all retained edges have zero length, so the previous branch covers every feasible target and the others may be unavailable. Otherwise apply Section 3 with its specified h_0. For each remaining target v, binary-search the first snapshot index τ with

$$U_\tau(v, 0, 0) - \rho \le 1/16. \tag{11}$$

If it exists, store its scale L_v and bound $B_v = (1 + 8\eta)L_v$; otherwise report unavailable.

To prove completeness, fix an original feasible target not covered by the zero branch. Let L_* be the optimum in the retained graph at budget C'. It is positive, attained by a simple path, and at most $\mathrm{OPT}_C(t)$. Its largest edge a lies in $[L_*/n, L_*]$. At L_* that edge's raw mean is strictly inside $[\delta, U]$. Take the largest grid

mean no greater than this raw mean. Its event satisfies

$$L_* \le L \le (1+\eta/2)L_* \le (1+\eta)\,\mathrm{OPT}_C(t). \tag{12}$$

Using the retained optimum here is essential: representatives may be much shorter than the original optimum.

At this event, all edges on that path are unblocked. On either clock, (8) bounds the total ideal Poisson mean by

$$(1+\eta/2)q/(1+4\eta) + n\delta \le q(1-2\eta).$$

For total mean at most $q(1-d)$, Chernoff's exponential-moment calculation gives tail at q at most $\exp(-q[-\log(1-d)-d]) \le \exp(-qd^2/2)$. With $d \ge 2\eta$ and (6), the two tails sum to less than $1/512$. Coupling compiled clocks to ideal clocks on at most n path edges adds at most $2nq\tau \le 2^{-19}$. Thus the compiled system can follow this fixed path with failure $f \le 1/512 + 2^{-19}$. By (4),

$$U(t,0,0) - \rho \le (16/15)(\rho+f) - \rho < 1/16.$$

The first accepted event therefore exists and has $L_t \le L$. This is an analysis using ideal random variables; the algorithm never samples them.

6 Actual witnesses with bounded numerical work

At a queried accepted scale, run one full rounded static solve on the retained graph and keep its pointers. This is query work, not a separate preprocessing run for each target. Its labels satisfy $X \le Z \le X + Nh_0$. Define $V = Z - \rho$, with source value zero and overflow value one. At the initial target,

$$V \le 1/16 + 2^{-20}, \qquad \mathbb{E}[V_{\mathrm{next}}] \le V$$

under the chosen pointer. Every pointer trajectory has at most N decisions.

Set $z = (1+4\eta)/(1+8\eta)$ and $d = 1-z$, and initialize two dyadic variables $H_\ell = H_c = 1/8$. Choose mesh $h = 2^{-b_d}$, with $b_d \ge 32 + 4\lceil \log_2(q+2)\rceil$ and $h \le 1/(2^{20}N)$. At a selected edge with rounded means μ_ℓ, μ_c, surviving increments (k,l) propose the analytic updates

$$H'_\ell = H_\ell e^{k\log z + d\mu_\ell}, \qquad H'_c = H_c e^{l\log z + d\mu_c}. \tag{13}$$

Give an overflow outcome potential one, omitting the H terms, and a surviving outcome potential $\Phi = V + H_\ell + H_c$. Equation (7) implies that the expected proposed potential is at most the current one. Independence lets one drop the other clock's survival restriction when bounding each H contribution. A tick multiplies the corresponding H by z.

The finite routine below computes an upper value for each proposed H with excess at most $2h$, or certifies it greater than two. Enumerate surviving increment pairs and choose the least upper score, omitting certified-large pairs. When $\Phi < 1$, some surviving outcome has true score at most Φ, since overflow has score one. Such an outcome is not omitted while $\Phi + 4h < 3/4$. A tick is rounded upward with excess at most h. Inductively,

$$\Phi \le 1/16 + 2^{-20} + 1/4 + 4Nh < 3/4. \tag{14}$$

Chosen outcomes never overflow and the pointer rank reaches the source within N decisions. Reverse the backward actual-edge walk, omit ticks, and cycle-erase with a vertex stack.

Upward enclosures preserve $H_\ell \ge (1/8)z^{\text{used length ticks}}\exp(d\sum \mu_\ell)$, including voluntary ticks, and similarly for cost. If physical length exceeded $(1+8\eta)L_t$, the raw-mean lower bound would give $\sum \mu_\ell > q/z$. Since fewer than q ticks were used,

$$H_\ell > (1/8)\exp\{q[\log z + d/z] - \log z\} > 1.$$

Indeed $\log(1-d)+d/(1-d)\ge d^2/2$ by its power series, $d\ge 8\eta/3$, and $q(8\eta)^2/18>\log 8$. This contradicts (14). The identical cost argument gives $c(P_t)\le(1+8\eta)C'$. Therefore (12) and the first-event choice give

$$B_t\le(1+8\eta)(1+\eta)\,\mathrm{OPT}_C(t)\le(1+\varepsilon)\,\mathrm{OPT}_C(t),\qquad c(P_t)\le(1+8\eta)(1+2\eta)C\le(1+\varepsilon)C.$$

Cycle erasure preserves both bounds. This also supplies a witness for any accepted originally infeasible target; an unreachable target cannot yield such a path. All represented paths satisfy their actual reported bounds.

6.1 Finite upper enclosures for the witness update

Here $b=b_d\ge 32$, $a=\lceil\log_2(q+2)\rceil$, and $0<d\le 1/4$. Form the exact rational sum

$$L_0=-\sum_{r=1}^{16(b+a+1)} d^r/r,\qquad x_0=kL_0+d\mu.$$

For $x=k\log z+d\mu$ and $k<q$, the geometric logarithm-tail bound gives $0\le x_0-x\le 2^{-32b}$. Stored H belongs to $[h,1]$. If $x_0\ge b+3$, then $He^x>2$; reject this component. If $x_0\le -b-3$, return h, an upper enclosure of excess at most h.

Otherwise choose a power of two R with $8(b+4)\le R<16(b+4)$, put $B=100(b+a+1)$, $h'=2^{-B}$, and $y=\lfloor x_0/R\rfloor_{h'}$. Thus $|y|<1/7$. Evaluate the degree $K=16b$ exponential polynomial by the rounded Horner recursion

$$A=1;\qquad A\leftarrow\lfloor 1+yA/r\rfloor_{h'}\quad(r=K,K-1,\dots,1).$$

Each Horner value is between zero and two. Rounding contributes less than $2h'$, rounding the argument less than $2h'$, and the Taylor tail at most $2\cdot 7^{-K-1}\le 2^{-32b-1}$. Hence

$$|A-e^{x_0/R}|\le 4h'+2^{-32b-1}.$$

Square $\log_2 R$ times, rounding down to h' each time. All intermediates are bounded by 2^{2b+16}. In relative error, a squaring step satisfies $r_j\le 2r_{j-1}+r_{j-1}^2+h'e^{b+3}$. For $r_j\le 2^{-b}$, the amplification over these steps is at most $2R$, since $(2+2^{-b})^{\log_2 R}\le 2R$. Using $e^{b+3}<2^{2b+6}$ and $R<2^b$ yields the loose absolute-error bound

$$2^{5b+32}(5h'+2^{-32b-1})\le 2^{-26b+1}.$$

The same estimates at earlier stages are smaller than 2^{-b} in relative error, verifying the bootstrap. Replacing x_0 by x adds at most $2^{2b+8}2^{-32b}$, so final exponential error is below 2^{-20b}. Add 2^{-10b} to obtain an upper enclosure, multiply by $H\le 1$, and round upward to h. The total excess is at most $2h$, as required in (14).

Every rounding search has an explicit bounded domain; negative arguments are shifted before binary search. The exact logarithm sum has $O(b+a)$ terms and polynomial rational encoding length. Later arithmetic uses $O(B+b)$-bit dyadics. In particular no unrounded rational power with exponent of order $1/\eta$ is stored, and neither exponential nor logarithm is an algorithmic primitive.

7 Operation count, storage, and scope

Put $p=1+\lceil\log_2(n+2)\rceil+\lceil\log_2(1/\eta+2)\rceil$. Then $q=O(\eta^{-2})$, $J,J_c=O(\eta^{-1}p)$, $M\le\min(m,n^2(J_c+1))$, $E\le MJ$, and $D=O(p)$. Cost buckets and the bounded-key map cost $O((m+n)p^2)$. Zero-length SSSP costs $O((M+n)p)$. Event creation and sorting cost $O(MJp)$. The shared compiler costs $O(J(q+p)p)$ operations and $O(Jq)$ storage.

DP labels use $b_0=O(p)$ bits. A joint-law candidate has at most q^2+1 terms: dyadic products and sums temporarily have $O(p+\log q)$ bits before rounding. Thus p^2 conservatively covers snapshot lookup,

rounding, and bounded numeric overhead in (5). The preprocessing bound is

$$O\big(E + (M+n)(1+q^2D^2)q^4[p^2 + \log(n+2)]\big).$$

Persistent storage is bounded as in Lemma 3, plus graph, shared probability tables, and $O(Mq^2)$ reused static scratch. Searching the threshold for all targets adds $O(np\log(n+2))$ work and $O(n)$ records, without n full static solves. Every term is at most $O((m+n)\eta^{-12}p^8)$. Absorbing powers of $\log(1/\eta)$ into extra powers of $1/\varepsilon$ gives the stated universal exponents 20, 100 after increasing an absolute constant.

A query performs one static solve on $M = \mathrm{poly}(n, 1/\varepsilon)$ edges, then at most N decisions with at most q^2 surviving candidates per edge decision and polynomial-p enclosure work per candidate. It is polynomial in the required parameters independently of all weight aspects. Static scratch, an $O(nq)$ output walk, and a cycle-erasure stack fit the larger preprocessing-space bound even while the persistent representation is retained.

For rational inputs, event values, original-edge comparisons, and SSSP sums have polynomial encoding length in the input and n, m. Grid indices and stored DP values use $O(p)$ bits. Rounding before reuse prevents denominator growth across recursive levels. The query's finite logarithm series has polynomial encoding length in the input encoding and p, followed by fixed-precision arithmetic. Control integers have $O(\log(n+m+1/\varepsilon))$ bits up to an absolute factor. This completes Theorem 1. If $C = 0$ is requested separately, delete positive-cost edges and solve ordinary length SSSP.

Limits and computational checks. The construction has intentionally large constants and accuracy exponents; no practical speedup is claimed. The established scope is the two-resource all-target theorem above. Exact rational tests exercised boundary values, closed classes, clipping, rounded Bellman residuals, finite ranks, and compiled CDFs on small instances. Separate high-precision decimal checks exercised the witness enclosure routine. These are diagnostic checks, not exhaustive executions at the prescribed q, formal verification, or proofs of the universal bounds; the arguments above supply the latter.

References

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AI methodology

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